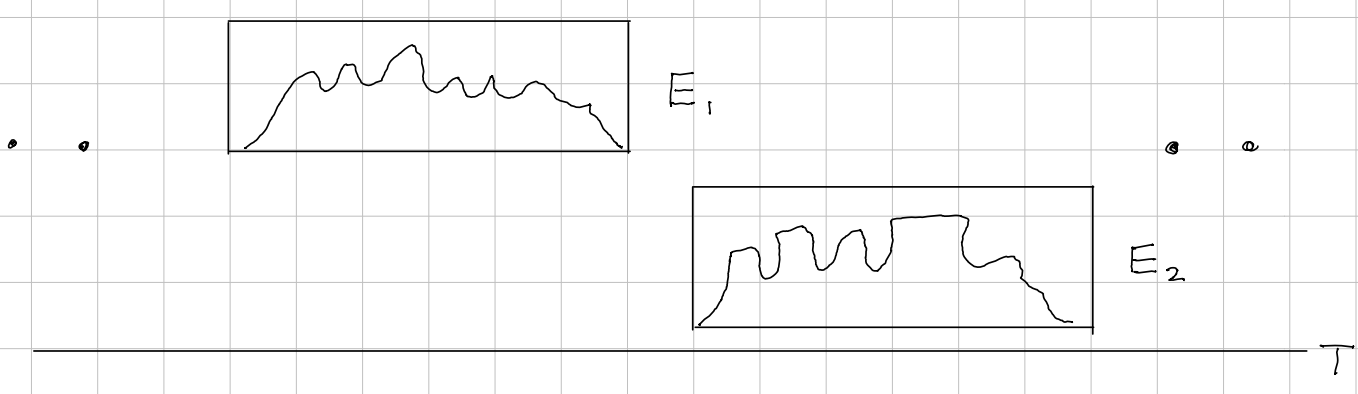
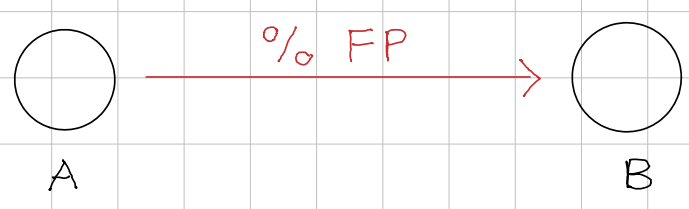
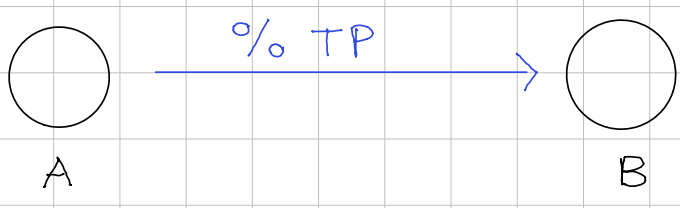


temporal co-occurrence: countersinging or coincidence?
- need to factor in false positives from various sources



* No vocalized interaction, may cause error to propagate



* interaction present & classified

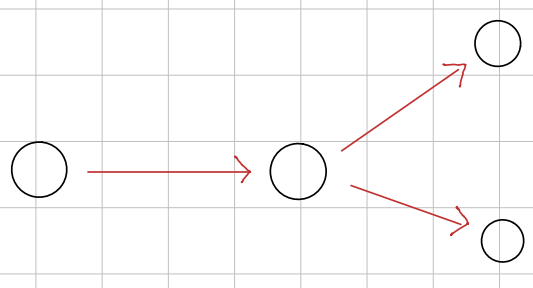
$$P(\text{precision}) = \frac{TP}{TP + FP}$$

* % correctly classified interactions
* optimize for this

* The focus of this study
* missing interactions

Actual \ Predicted	Yes	No
Yes	TP	FN
No	FP	TN

* Most detrimental, need to account for
* Not available from the data



* FP's compound across a bird network, propagating loss

So, we need to consider errors (FP, FN) within the model.

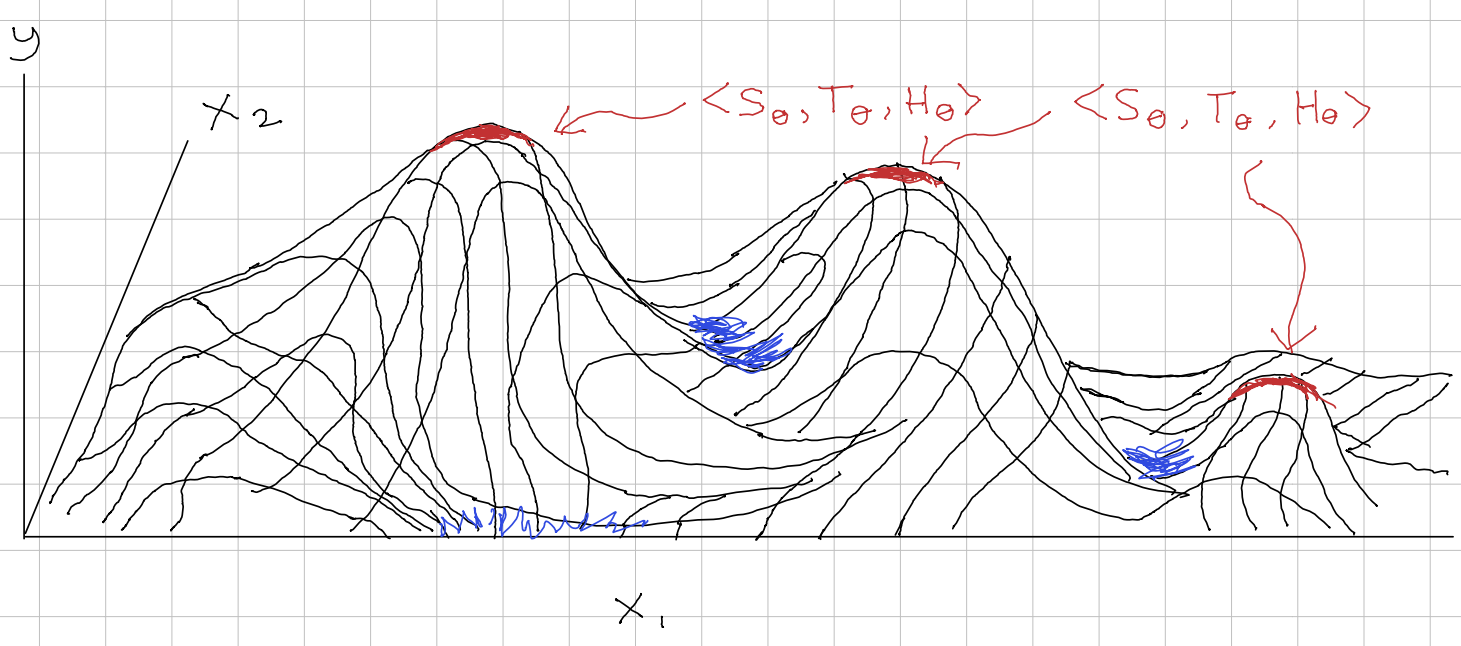
∴ Developing a framework to navigate bias-var

What is the computational budget for accuracy & computation?

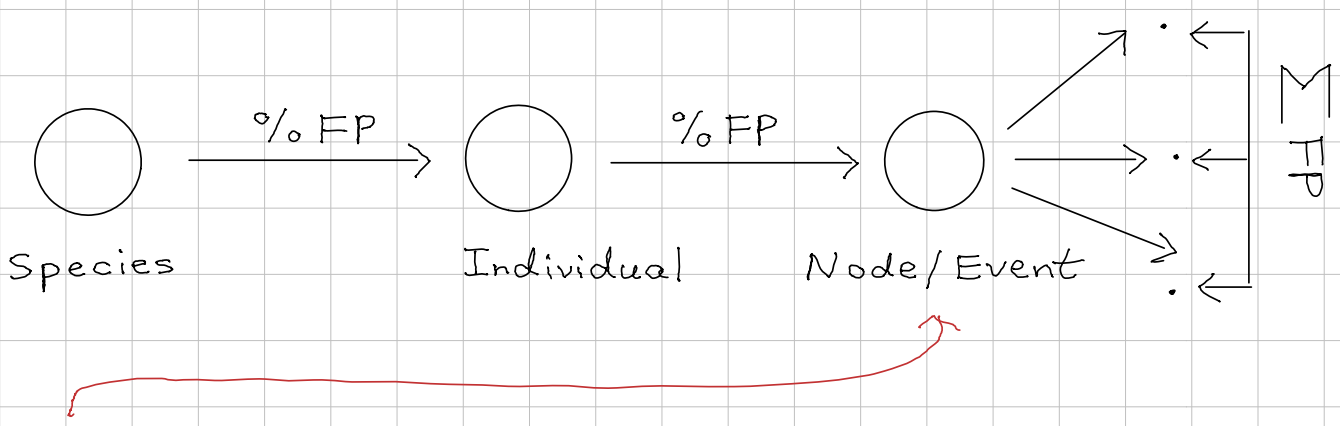
Gaussian Process Models

- will need a way to identify high confidence interactions

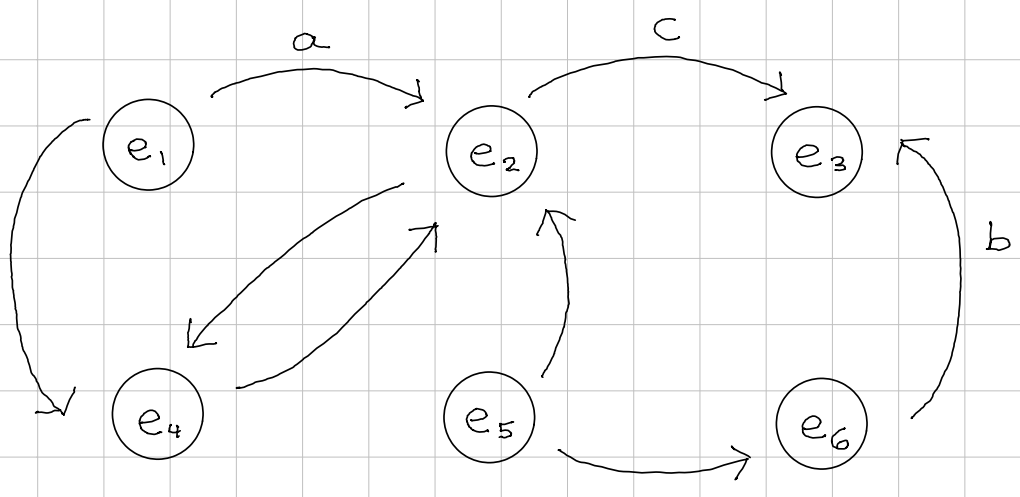
* Gaussian Process Model take in factors
S, T, H ⟶ cont. dist of $p(\vec{f})$



Acoustic interactions may contain covariance among features — $Cov(S,T) \neq 0$ & may not be independent — $p(S,T,H) \neq p(S)p(T)p(H)$

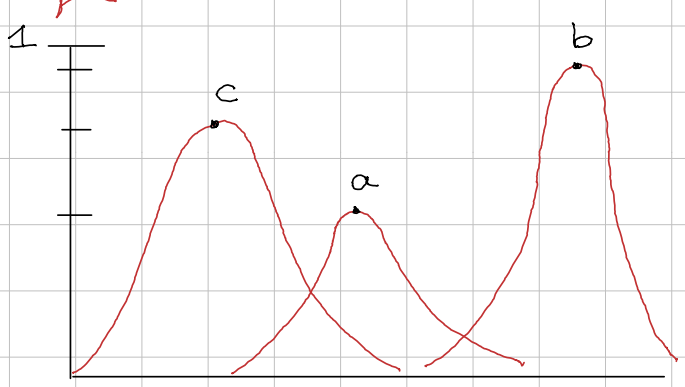


What I need to account for in network estimated from the data



Each edge will have uncertainty quantification applied

$p(\text{Interaction} \mid S, T, H)$



- Confidence will be according to probability distribution

← more confident data

$B = S, T, H$

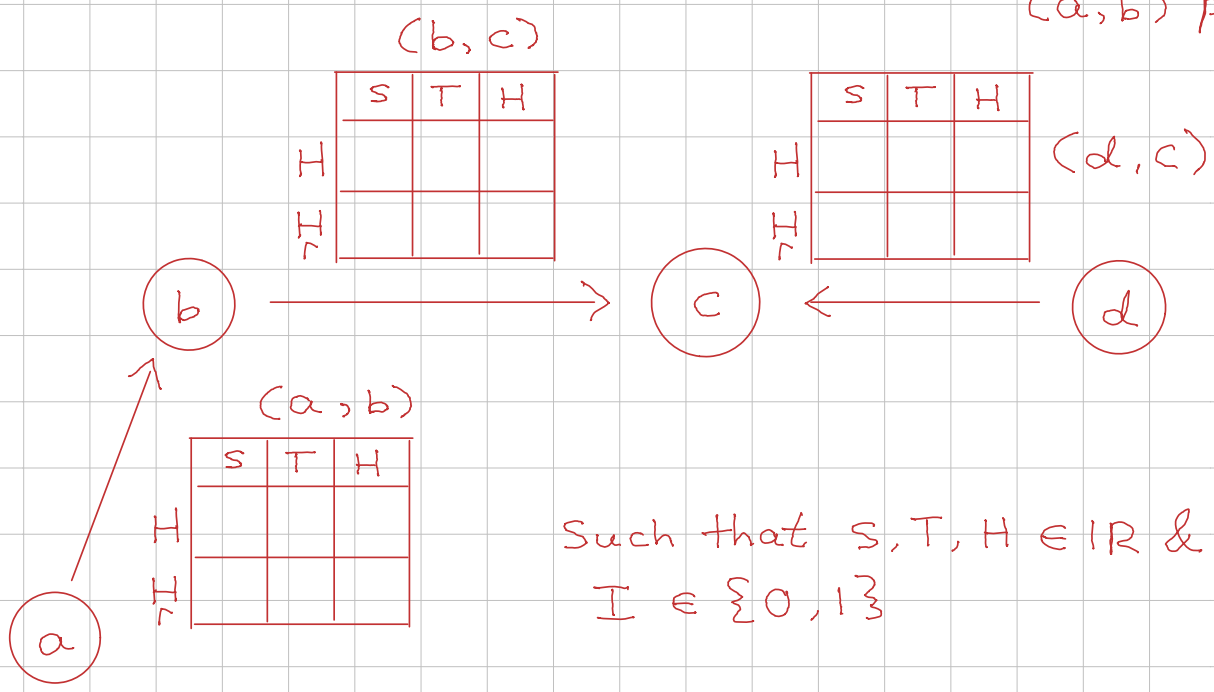
$$P(\text{Interaction}) = \iiint_{S\,T\,H} P(\text{Interaction} \mid S, T, H)\, ds\, dt\, dh$$

* avg. int. across all edges

$$= \iiint_{S\,T\,H} \frac{P(\text{Interaction}, S, T, H)}{P(S, T, H)}\, ds\, dt\, dh$$

* all combinations

$P_{ab}(\text{Interaction} \mid S = s_0, T = t_0, H = h_0)$ * Conditional for (a, b) pair



Such that $S, T, H \in \mathbb{R}$ & $I \in \{0, 1\}$

* Combinatorial explosion is a risk with adding features → 3 explainable features is fine